The University of Sydney

MOOC Introduction to Calculus

Notes for 'The Chain Rule'

Important Ideas and Useful Facts:

(i) Differentiable functions: A function is called *differentiable* if its graph is smooth and the derivative exists at all points of interest. The graphs of differentiable functions are continuous, that is, able to be drawn without lifting pen off paper (roughly speaking and in your imagination). However functions with continuous graphs need not be differentiable (for example, if there are sharp points or cusps).

(ii) The Chain Rule: If f and g are differentiable then $f \circ g$ is differentiable whenever it is sensibly defined at x, in which case

$$(f \circ g)'(x) \;=\; f'(g(x))g'(x) \;,$$

or equivalently, using Leibniz notation, where we put $y = f(u)$ and $u = g(x)$,

$$\frac{dy}{dx} \;=\; \frac{dy}{du}\,\frac{du}{dx} \;,$$

so that we may think of the differentials du in the 'numerator' and 'denominator' as 'cancelling'.

Examples and derivations:

1. Find $\dfrac{dy}{dx}$ when $y = (3x-7)^4$.

Solution: Put $u = 3x - 7$, so that $\dfrac{du}{dx} = 3$, and $y = u^4$, so that $\dfrac{dy}{du} = 4u^3$. Hence

$$\frac{dy}{dx} \;=\; \frac{dy}{du}\,\frac{du}{dx} \;=\; 4u^3(3) \;=\; 12u^3 \;=\; 12(3x-7)^3 \;.$$

2. Find $\dfrac{dy}{dx}$ when $y = (2x^2 + 3x + 1)^3$.

Solution: Put $u = 2x^2 + 3x + 1$, so that $\dfrac{du}{dx} = 4x + 3$, and $y = u^3$, so that $\dfrac{dy}{du} = 3u^2$. Hence

$$\frac{dy}{dx} \;=\; \frac{dy}{du}\,\frac{du}{dx} \;=\; 3u^2(4x+3) \;=\; 3(2x^2+3x+1)^2(4x+3) \;.$$

3. Find $\dfrac{dy}{dt}$ when $y = e^{-5t}$.

Solution: Put $u = -5t$, so that $\dfrac{du}{dt} = -5$, and $y = e^u$, so that $\dfrac{dy}{du} = e^u$. Hence

$$\frac{dy}{dt} \;=\; \frac{dy}{du}\,\frac{du}{dt} \;=\; e^u(-5) \;=\; -5e^{-5t} \;.$$

4. Find $\dfrac{dy}{dx}$ when $y = e^{x^2}$.

Solution: Put $u = x^2$, so that $\dfrac{du}{dx} = 2x$, and $y = e^u$, so that $\dfrac{dy}{du} = e^u$. Hence

$$\frac{dy}{dx} \;=\; \frac{dy}{du}\,\frac{du}{dx} \;=\; e^u(2x) \;=\; 2xe^{x^2}\ .$$

5. Find $\dfrac{du}{d\theta}$ when $u = \sin(6\theta)$.

Solution: Put $v = 6\theta$, so that $\dfrac{dv}{d\theta} = 6$, and $u = \sin v$, so that $\dfrac{du}{dv} = \cos v$. Hence

$$\frac{du}{d\theta} \;=\; \frac{du}{dv}\,\frac{dv}{d\theta} \;=\; (\cos v)(6) \;=\; 6\cos(6\theta)\ .$$

6. Find $\dfrac{dy}{dx}$ when $y = \cos(\sin x)$.

Solution: Put $u = \sin x$, so that $\dfrac{du}{dx} = \cos x$, and $y = \cos u$, so that $\dfrac{dy}{du} = -\sin u$. Hence

$$\frac{dy}{dx} \;=\; \frac{dy}{du}\,\frac{du}{dx} \;=\; (-\sin u)(\cos x) \;=\; -\cos x\sin(\sin x)\ .$$

7. Find $\dfrac{dx}{dt}$ when $x = e^{\cos(5t)}$.

Solution: Put $u = \cos(5t)$ and $v = 5t$, so that $\dfrac{dv}{dt} = 5$, and $u = \cos v$, so that $\dfrac{du}{dv} = -\sin v$. Also $x = e^u$, so that $\dfrac{dx}{du} = e^u$. Hence

$$\frac{dx}{dt} \;=\; \frac{dx}{du}\,\frac{du}{dv}\,\frac{dv}{dt} \;=\; (e^u)(-\sin v)(5) \;=\; -5e^{\cos(5t)}\sin(5t)\ .$$

8. Let f and g be functions such that $f(x) = x^3$ and $g(x) = x^2+1$. Find $(f\circ g)'(x)$ and $(g\circ f)'(x)$.

Solution: Observe that $f'(x) = 3x^2$ and $g'(x) = 2x$, so that

$$(f\circ g)'(x) \;=\; f'(g(x))g'(x) \;=\; 3(x^2+1)^2(2x) \;=\; 6x(x^2+1)^2\ ,$$

and

$$(g\circ f)'(x) \;=\; g'(f(x))f'(x) \;=\; 2x^3(3x^2) \;=\; 6x^5\ .$$

9. Let f and g be functions such that $f(x) = x^2+x+1$ and $g(x) = (x-1)^3$. Find $(f\circ g)'(x)$ and $(g\circ f)'(x)$.

Solution: Observe that $f'(x) = 2x+1$ and $g'(x) = 3(x-1)^2$, so that

$$(f\circ g)'(x) \;=\; f'(g(x))g'(x) \;=\; (2(x-1)^3+1)(3)(x-1)^2 \;=\; 3(x-1)^2\big(2(x-1)^3+1\big)\ ,$$

and

$$(g\circ f)'(x) \;=\; g'(f(x))f'(x) \;=\; 3(x^2+x)^2(2x+1) \;=\; 3x^2(x+1)^2(2x+1)\ .$$

10. We sketch a proof that of the Chain Rule, using Leibniz notation. The claim is that

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx},$$

where $y = f(u)$ is a function of u and $u = g(x)$ is a function of x.

Sketch of proof: By the limit definition of derivative, expressed in terms of small changes in the variables, we have, assuming that we can choose Δu always to be nonzero,

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \left(\frac{\Delta y\, \Delta u}{\Delta u\, \Delta x} \right) = \left(\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta u} \right) \left(\lim_{\Delta x \to 0} \frac{\Delta u}{\Delta x} \right)$$

since the limit of a product is the product of the limits. But observe that

$$\left(\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta u} \right) = \left(\lim_{\Delta u \to 0} \frac{\Delta y}{\Delta u} \right)$$

because Δu goes to zero as Δx goes to zero (since there is a blanket assumption that all functions are differentiable, and hence continuous, a subtle point elaborated in more advanced courses in mathematics). Hence, we get

$$\frac{dy}{dx} = \left(\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta u} \right) \left(\lim_{\Delta x \to 0} \frac{\Delta u}{\Delta x} \right) = \left(\lim_{\Delta u \to 0} \frac{\Delta y}{\Delta u} \right) \left(\lim_{\Delta x \to 0} \frac{\Delta u}{\Delta x} \right) = \frac{dy}{du}\frac{du}{dx},$$

at the last step, again by the limit definition of the respective derivatives, and the sketch of the proof is complete.